QUBO Formulation: 0/1 Knapsack Problem

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Problem Statement. Given N items, each characterized by a value v_i and a weight w_i for $i \in \{0, \ldots, N-1\}$, and a knapsack with a maximum weight capacity W, the goal is to select a subset of items to maximize the total value without exceeding the capacity W.

Decision Variables. For each item $i \in \{0, \ldots, N-1\}$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i takes the value 1 if item i is included in the knapsack, and 0 otherwise.

QUBO Derivation. Step 1. The original optimization problem seeks to maximize $\sum_{i=0}^{N-1} v_i x_i$. Since QUBO solvers minimize quadratic functions, we negate the objective to obtain the minimization form:

$$\min_{x \in \{0,1\}^N} \left(-\sum_{i=0}^{N-1} v_i x_i \right). \tag{1}$$

Step 2. The weight capacity constraint is expressed as:

$$\sum_{i=0}^{N-1} w_i x_i \leq W. \tag{2}$$

Step 3. We enforce the constraint via a quadratic penalty term with weight $\lambda > 0$. To convert the inequality into a penalty that activates upon violation, we write:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} w_i x_i - W \right)^2. \tag{3}$$

Expanding the square yields:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} w_i^2 x_i^2 - 2W \sum_{i=0}^{N-1} w_i x_i + W^2 + \sum_{i=0}^{N-1} \sum_{j \neq i} w_i w_j x_i x_j \right). \tag{4}$$

Using the idempotent property of binary variables, $x_i^2 = x_i$, the linear terms simplify. Additionally, we symmetrize the double sum by restricting to $i < j$ and multiplying by 2:

$$P(x) = \lambda \left(\sum_{i=0}^{N-1} (w_i^2 - 2Ww_i)x_i + \sum_{0 \le i < j \le N-1} 2w_i w_j x_i x_j + W^2 \right). \tag{5}$$

Step 4. Combining the negated objective and the penalty term gives the full QUBO Hamiltonian $H(x)$:

$$H(x) = -\sum_{i=0}^{N-1} v_i x_i + \lambda \sum_{i=0}^{N-1} (w_i^2 - 2Ww_i)x_i + \lambda \sum_{0 \le i < j \le N-1} 2w_i w_j x_i x_j + \lambda W^2. \tag{6}$$

Step 5. Reading off the coefficients from $H(x) = \sum_{i \le j} Q_{i,j} x_i x_j + c$, the QUBO matrix entries and offset are given in closed form as:

$$Q_{i,i} = -v_i + \lambda(w_i^2 - 2Ww_i), \quad \text{for } i \in \{0, \ldots, N-1\}, \tag{7}$$

$$Q_{i,j} = 2\lambda w_i w_j, \quad \text{for } 0 \le i < j \le N-1, \tag{8}$$

$$Q_{i,j} = 0, \quad \text{for } i > j, \tag{9}$$

$$c = \lambda W^2. \tag{10}$$

Penalty Weight Justification. The penalty parameter λ must be chosen sufficiently large to ensure that constraint violations are strictly penalized over any potential gain in the objective function. A sufficient condition is $\lambda > \sum_{i=0}^{N-1} v_i$. This guarantees that the minimum penalty incurred by any infeasible assignment exceeds the maximum possible total value achievable by any feasible subset, thereby forcing the global minimum of $H(x)$ to reside within the feasible region.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal solution to the original knapsack problem. For any feasible solution, the constraint is satisfied, the penalty term vanishes, and $H(x)$ reduces exactly to the negated objective value. For any infeasible solution, the penalty term is strictly positive. By the choice of λ, the energy increase from violating the constraint outweighs any possible decrease in the objective term, ensuring that the energy of every infeasible point is strictly greater than that of at least one feasible point. Consequently, the global minimum of $H(x)$ must correspond to a feasible solution that maximizes the original objective.

Illustrative Example. Consider a concrete instance with $N = 3$ items, values $v = [10, 20, 30]$, weights $w = [5, 10, 15]$, capacity $W = 20$, and penalty weight $\lambda = 100$. Substituting these into the general formulation yields the following components:

The concrete minimization objective is $\min -10x_0 - 20x_1 - 30x_2$. The concrete penalty term is $100(5x_0 + 10x_1 + 15x_2 - 20)^2$. Expanding the penalty using the general derivation

gives:

$$P(x) = 100\left[(25-200)x_0 + (100-400)x_1 + (225-600)x_2 + 2(50x_0x_1 + 75x_0x_2 + 150x_1x_2) + 400\right]. \tag{11}$$

Combining objective and penalty yields the resulting Hamiltonian:

$$H(x) = -1810x_0 - 30200x_1 - 37530x_2 + 10000x_0x_1 + 15000x_0x_2 + 30000x_1x_2 + 40000. \tag{12}$$

Evaluating $H(x)$ over all $2^3 = 8$ binary assignments reveals that the feasible solutions (total weight ≤ 20) are $x \in \{(0,0,0),(1,0,0),(0,1,0),(0,0,1),(1,1,0)\}$. Among these, the assignment $x^* = (0,0,1)$ achieves the maximum value of 30 and yields the minimum energy:

$$H(x^*) = -37530 + 40000 = 2470. \tag{13}$$

Thus, $x^* = (0,0,1)$ is the optimal solution with energy 2470.